A Key Tool!

The chain rule:

$$\frac{d}{dx}(f(g(x))) = _____.$$

Implicit Differentiation

- Consider the equation $xy + y = e^x$. This defines y as function of x. Find $y(x)$ and use it to find $\frac{dy}{dx}$.

Now, consider the equation $x^2 + y^2 = 4$. What curve does this equation describe? Draw it!

The equation $x^2 + y^2 = 4$ does not define y as a function of x. Why not?

However, "near the point $(0, 2)$", y is function of a x. Highlight such a region on your graph above. What is this function? What is the derivative of this function at $x = 1$?

Do the same for the point $(0, -2)$ (and $x = -1$).

Notice that in both the above cases, we can write $\frac{dy}{dx} = _____$.

Even though $x^2 + y^2 = 4$ does not define y as a function of x, it is still close to a function that the derivative makes sense, and the usual rules apply. Let's pretend y is a function of x, differentiate both sides, and solve for $\frac{dy}{dx}$:

Notice that $\frac{dy}{dx}$ depends on both x and y. For "normal" functions, the derivative should only depend on the independent variable. Why isn't this true for our equation?

Are there any points where $\frac{dy}{dx}$ is not defined? What happens to the tangent line to the curve at these points?

- Let's revisit $xy + y = e^x$. What is $\frac{dy}{dx}$? Check that it's the same as what you got previously!

Implicit Differentiation

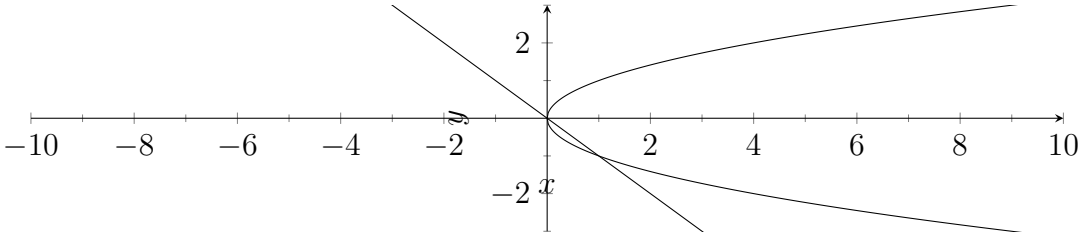
What we carried out in these cases is a process of *implicit differentiation*: We *implicitly* assume that y is a function of x, making expressions like y^2, e^y, etc *compound* functions: $y^2 = [y(x)]^2$, $e^y = e^{y(x)}$, etc. We then use the chain rule to differentiate them.

Questions

1. (a) Find the tangent line to $x^2 + xy - y^3 = xy^2$ at the point $(1, 1)$.

☐Regrade

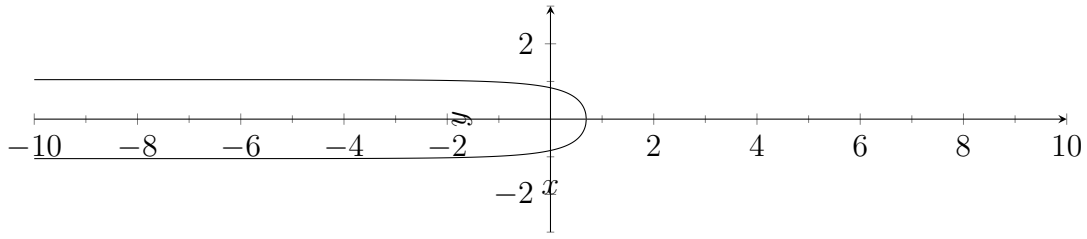
(b) Here's a picture of this curve. Odd, eh? Plug the point $(0, 0)$ and $(1, -1)$ into your expression for $\frac{dy}{dx}$? What do you get? Looking at the graph, what can you say about the slope at those points? More on this in lab...



2. (a) Find $\frac{dy}{dx}$ if $e^x + e^{y^2} = 3$.

☐Regrade

This curve looks like this:

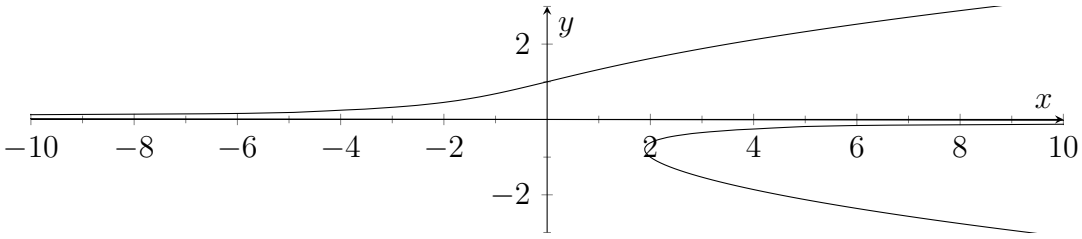


(b) What is $\frac{dy}{dx}$ when $y = 0$? Find such a point on the curve. What can you say about the direction of the tangent line there?

(c) As x approaches $-\infty$, what does $\frac{dy}{dx}$ approach? Make sure you see how this matches up with the graph!

3. Consider the curve defined by $y^3 - xy = 1$:

☐Regrade



(a) Find $\frac{dy}{dx}$

(b) As we saw in the previous question, curves have vertical tangent lines when their derivative is undefined. That is, when the denominator is zero. In the case of this curve here, this is true when $x =$ _____. Plug this expression for x into the formula for the curve above and solve to find the y coordinate of this point. Then find the x coordinate. Check on the curve above that there's indeed a vertical tangent line at that point.

(c) Curves have horizontal tangent lines when $\frac{dy}{dx} =$ _____. Carry out a similar process to (b) to find out where (if anywhere) this curve has a horizontal tangent line.